

Shaik Abdus Sattar

Software Engineer – AI/ML

+91-7032698579 shaik.abdus.sattar@outlook.com

linkedin.com/in/seventie github.com/seventie shaikabdussattar.vercel.app

Education

Amrita Vishwa Vidyapeetham

Bachelor of Engineering in Computer Science; CGPA: 8.4/10.0

Coimbatore, Tamil Nadu

Aug 2023 – Present

Excellencia Junior College

Intermediate/+2 (State Board): 98%; JEE-Mains: 95.87 percentile

Hyderabad, Telangana

2023

Technical Skills

Machine Learning & AI: Deep Learning, CNNs, Transformers, NLP, Computer Vision

ML Frameworks & Data: PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn

Programming Languages: Python, C++, Java, SQL, JavaScript, Haskell

Full-Stack Development: Node.js, React, Next.js, Tailwind CSS, Express.js, MongoDB

Research Experience

Clinically-Informed Deep Learning for Radiotherapy QA Prediction

2025 – Present

Research Assistant | Advisors: [Dr. Amit Agarwal](#), [Dr. Rose Kamal](#)

- Automated pre-treatment Gamma-passing-rate prediction for radiotherapy QA, benchmarked against dose-map and complexity-metric ground truth, by building CNN/hybrid deep-learning models with clinically weighted loss functions.

Efficient 3D Molecular Graph Kernels for Molecular Property Prediction

2026 – Present

Research Assistant | Advisor: [Dr. T. Ramraj](#)

- Designing optimal data structures and a hierarchical 2D-to-3D filtering strategy to reduce the computational overhead of 3D molecular graph kernel construction for large-scale property prediction.
- Improved molecular similarity modeling for large-scale property prediction, validated via bond-length, bond-angle, and torsional feature comparisons, by designing scalable 2D-to-3D kernel frameworks.

Projects

Aurova: Mental Health Platform | RAG, LLMs, Full-Stack



- Shipped a privacy-first mental-health platform with anonymous journaling and peer support, secured via encrypted authentication, by building the full-stack app end-to-end.
- Reduced unsafe-response risk in AI counseling, verified via built-in crisis detection, by building a RAG-based LLM pipeline with context-aware personalization.

LFW Face Generation & Latent Space Editing | AutoEncoders, VAE, Latent Interpolation, PyTorch



- Synthesized photorealistic faces from the LFW dataset, evaluated via smooth latent-space interpolation, by training VAE models end-to-end in PyTorch.
- Isolated controllable facial attributes, confirmed via disentanglement analysis of the latent space, by implementing attribute-driven latent vector editing.

Contradictory, My Dear Watson – Cross-Lingual NLI | XLM-RoBERTa, Transformers, NLP



- Fine-tuned a pretrained XLM-RoBERTa transformer end-to-end for three-class entailment classification across 15+ languages, by designing custom tokenization and sequence-formatting pipelines for claim-evidence pairs.
- Outperformed baseline ML models on cross-lingual entailment classification, benchmarked on held-out multilingual test sets, by fine-tuning the full transformer rather than relying on frozen embeddings.

Certifications

AWS Cloud Practitioner Essentials – AWS (2025)

Deep Learning Specialization – DeepLearning.ai (2025)

Machine Learning Specialization – Stanford Online (2025)

Achievements & Leadership

Head Coordinator – Gokulashtami Event (2025)

Hackathon Winner – IETE Hackathon (2025)

AIR 270 – AEEE Exam (2023)

1st Place – Western Group Music, Gokulashtami (2025)